

Part A · Multiple Choice

2 points each

1. **d) Nicolaus Copernicus**

p. 67 — a radical idea at the time. It gained acceptance slowly, and our modern view of solar system motion is based on his work.

2. **b) the 23.5° tilt of Earth's axis combined with Earth's revolution around the Sun**

p. 70 — the text names choice (a) explicitly as a common misconception. The tilt means one hemisphere is more directly exposed to the Sun's rays than the other.

3. **a) the Tropic of Cancer, 23.5° N**

p. 70 — around June 21st or 22nd. At that same moment it is the winter solstice in the Southern Hemisphere.

4. **c) UVC**

pp. 68–69 — UVC is the highest energy and is stopped completely. UVB is mostly stopped; UVA, the lowest energy, travels through to the ground. Ozone does most of this filtering.

5. **b) 23 times as much heat**

p. 70 — 10,600 times is CFC-12. Carbon dioxide still matters enormously because it is far more abundant in the atmosphere than either.

6. **a) the Sun's rays shine most directly at the equator and daylight and nighttime hours are equal**

p. 71 — the equinoxes fall halfway between the solstices: the vernal equinox around March 21st or 22nd and the autumnal equinox around September 22nd or 23rd.

Part B · True or False

1 point each

7. **TRUE**

p. 68 — only about 7% is ultraviolet; the remainder is infrared, the longest wavelength, which we feel as heat.

8. **FALSE**

All of the greenhouse gases listed — CO₂, H₂O, methane, O₃, and nitrous oxides — are a normal part of the atmosphere. CFCs are the single exception. Human activity has, however, sharply raised the levels of several of them. (p. 70)

Part C · Short Answer

3 points each

9. **In June the North Pole is tilted toward the Sun, so the rays strike more directly and the days are longer; in December the tilt points away, spreading the same light over a larger area for fewer hours.**

Rubric: 1 pt for the tilt toward/away from the Sun. 1 pt for directness — when the tilt points away, light is spread out over a larger area, so that area is not heated as much. 1 pt for duration — fewer daylight hours in winter means less time for the Sun to warm the surface. Deduct if the student attributes the difference to Earth's distance from the Sun. (pp. 69–71)

10. A diagram showing the Sun's position in the sky, viewed from a fixed place on Earth at the same mean solar time, as it varies over a year. The shape comes from Earth's axial tilt and its orbital eccentricity.

Rubric: 1.5 pts for the definition — a plot of the Sun's position from one fixed location at the same clock time across a full year. 1.5 pts for both causes: axial tilt (obliquity), which produces the north–south component, and the eccentricity of Earth's orbit, which produces the east–west component (the equation of time). Full credit if the student notes that with a circular orbit and no tilt the analemma would be a single dot. (pp. 72–74)